SEONGGU JEONG · 22200667

Optimal Depth of Multimodal Fusion and Mitigating Modality Collapse

Investigating where visual features should be fused in a multimodal translation decoder — and why even carefully engineered solutions still fail to make models truly "see."

INTRODUCTION

Why Vision Is Necessary for Translation

The Textual Ambiguity Problem

Standard MT cannot resolve gender, color, or referential ambiguity from text alone — the image is required.

Multimodal MT (MMT)

Integrates visual features alongside text to disambiguate at translation time.

Modality Collapse

Models learn to ignore visual input entirely, relying on language priors alone.

The Evaluation Gap

BLEU cannot verify visual reasoning. **CoMMuTE** evaluates true visual disambiguation capability.

BACKGROUND

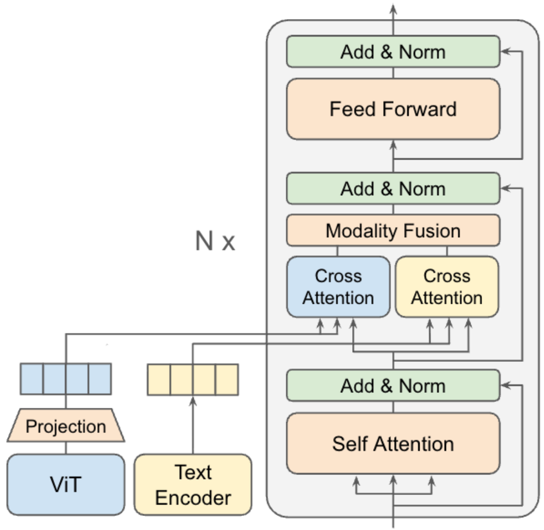
The Dual Attention Decoder (DAD) Baseline

Architecture

Independent cross-attention mechanisms for text and image within the decoder, combined via static fusion:

$$\text{Output} = 0.5 \times \text{Text} + 0.5 \times \text{Image}$$

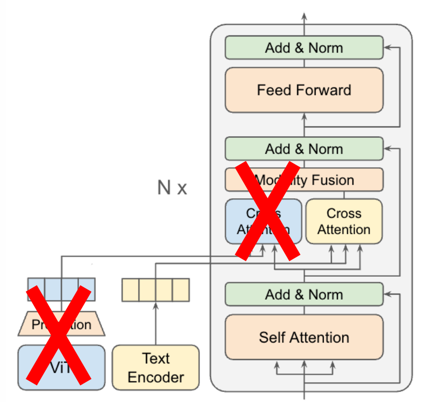
Dynamic gating was extensively tested — the fixed 0.5_sum proved most robust, establishing a strong baseline.



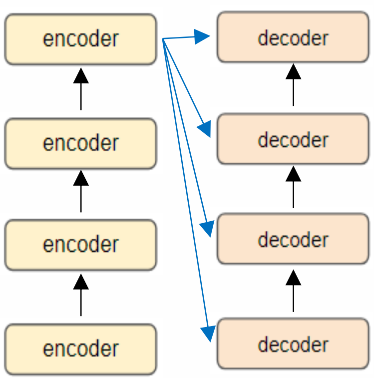
PROPOSED METHOD 1

Searching for Optimal Fusion Depth

Hypothesis: Can modality collapse be mitigated by changing *where* cross-attention occurs in the decoder?

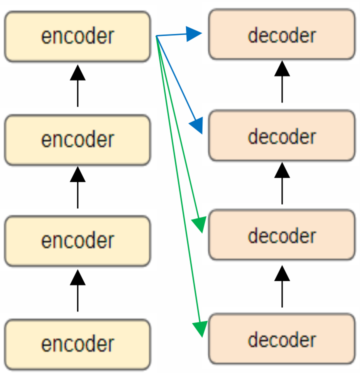


Text only case



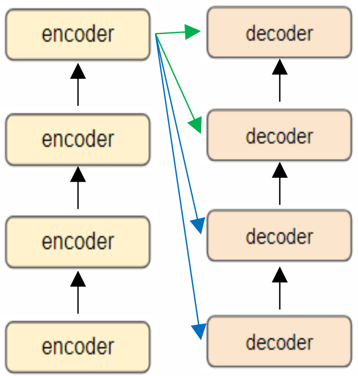
All-layers (DAD)

Fusion across every decoder layer to balance grounding and fluency



Bottom-heavy

Fusion at early decoder layers for strong visual grounding



Top-heavy

Fusion at later decoder layers to preserve translation fluency

Objective: Identify the optimal depth that maximizes visual grounding without degrading translation quality.

Text+ image →
Text only →

PROPOSED METHOD 2

Text-Guided Attention Pooling

The Alignment Challenge

To align the semantic spaces, we must match the text with the image. However, the Vision Transformer (ViT) outputs 196 separate image patches. We need a way to compress these into a single global visual representation.

The Problem with Naive Compression

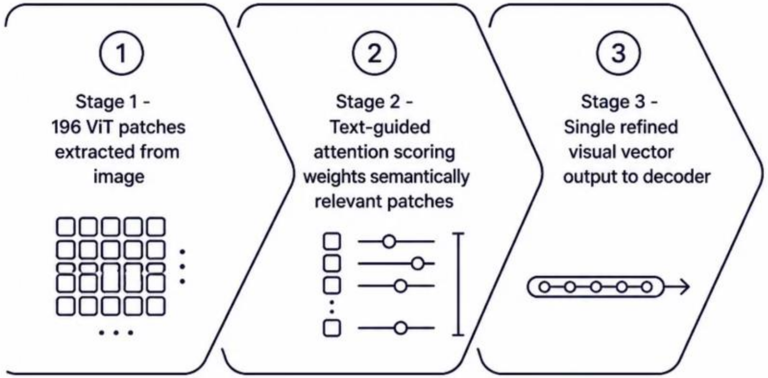
Averaging all 196 ViT patches destroys fine-grained detail — a red shirt blends into a blue sky.

The Solution: Text as Query

Instead of blind averaging, we use the global text vector as a "magnifying glass" over the 196 visual patches before they reach the decoder.

Selective Extraction

By calculating attention scores, we apply high weights only to patches semantically relevant to the text query, outputting a single, highly refined visual vector.



PROPOSED METHOD 3

InfoNCE Alignment & the Information Bottleneck

$$L_{\text{InfoNCE}} = -\mathbb{E}\left[\log \frac{\exp\left(\text{sim}\left(\mathbf{x},\mathbf{x}^{+}\right)/\tau\right)}{\sum_{\mathbf{x}^{-}\in\mathcal{N}} \exp(\text{sim}(\mathbf{x},\mathbf{x}^{-})/\tau)}\right]$$

InfoNCE Auxiliary Loss

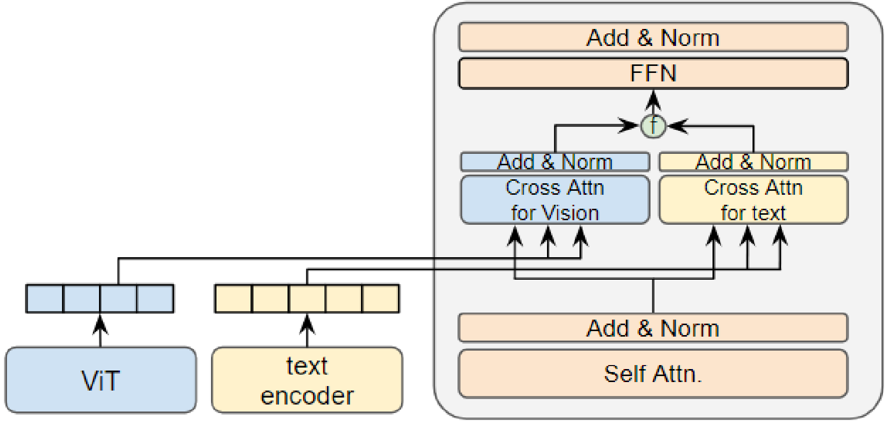
Applied at encoder level, forcing the pooled visual vector to align with its corresponding text vector in semantic space.

The "Lazy Decoder" Problem

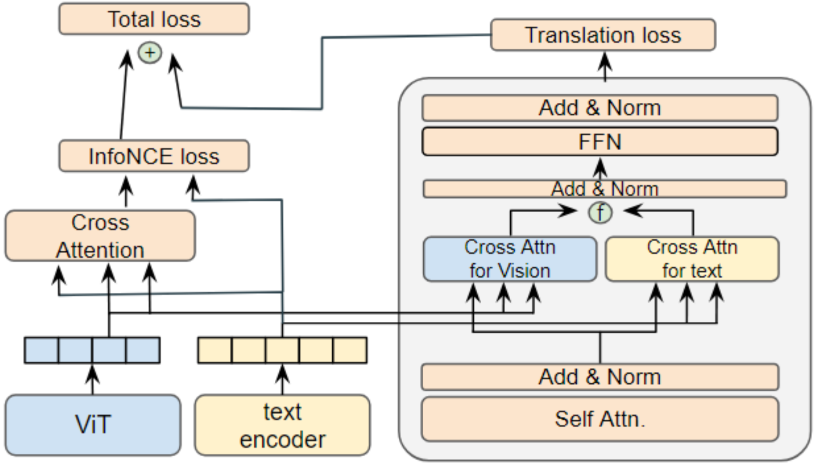
Even with aligned visual data, the decoder ignores it if source text is complete. A perfectly aligned image is useless if the decoder never looks.

PROPOSED METHOD 3

Text-Guided Attention Pooling+ InfoNCE Alignment



DAD



Proposed Model (InfoNCE)

EXPERIMENTAL SETUP

Datasets, Models & Evaluation

Datasets

Multi30k (in-domain) and **MSCOCO** (out-of-domain) · English → German

Vision Encoder

ViT-Tiny (16×16 patches). ViT-Base tested — marginal gains at disproportionate parameter cost.

Baseline

DAD model with 0.5_sum static fusion

Three Evaluation Tests

01

Test 1: BLEU on Multi30k

General in-domain translation quality

02

Test 2: BLEU on MSCOCO

Out-of-domain generalization

03

Test 3: CoMMuTE

Visual disambiguation — the true measure of whether the model "sees"

RESULTS 1

The Depth Illusion

	test1	test2	test3
all	40.95	29.45	50.22
bottom_heavy	41.29	29.27	49.78
top_heavy	41.64	29.35	50.56
text1dual3	41.53	29.61	49.89
text3dual1	41.15	29.82	50.00

The Depth Search Experiments

*All results are from our own experiments, averaged over three random seeds.

The CoMMuTE Paradox

Across **all** fusion depth variants, CoMMuTE scores remained nearly **50.00%** — random chance in a binary forced-choice test. General BLEU was competitive (~41.6) regardless of depth, but the model was **fundamentally blind**.

⚠ Changing *where* fusion occurs does not solve Modality Collapse. Semantic spaces must be aligned *first*.

RESULTS 2 & 3

Translation Quality Preserved — Collapse Persists

method	fusion_depth	test1	test2	test3
baseline (DAD)	all	40.95	29.45	50.22
baseline	top_heavy	41.64	29.35	50.56
CLS	all	41.40	29.93	50.00
infoNCE	all	41.56	29.42	50.00
infoNCE	top_heavy	41.69	29.64	49.78

Experiments on CLS, infoNCE

Improved Translation Quality (BLEU)

The proposed InfoNCE (top_heavy) achieved the highest BLEU score (41.69).
Demonstrates that semantic alignment slightly enhances general language modeling.

Persistent Modality Collapse (CoMMuTE)

Scores remained strictly stagnant at the ~50% baseline.
Proves that encoder-level alignment alone is insufficient to break the decoder's strong text bias.

*All results are from our own experiments, averaged over three random seeds.

DISCUSSION

Discussion & Failure Analysis

Despite our efforts to integrate visual features, modality collapse persisted. Here's why our current solutions fell short:

Limits of Encoder Alignment

InfoNCE aligned textual and visual semantic spaces, showing slight BLEU improvements. Yet, CoMMuTE scores remained at a random 50%. This demonstrates that merely providing high-quality, aligned visual features isn't enough to overcome the model's reliance on language priors.

The Trap of Rigid Fusion

The 0.5_sum fusion method, while stable for general translation, proved too inflexible. Forced to weigh modalities equally, the decoder optimized for safety by completely nullifying visual weights, avoiding potential noise during easy, text-only translations.

Significance

Our findings confirm that encoder-level alignment alone is insufficient. To truly combat modality collapse, semantic alignment must be complemented by a dynamic, decoder-level fusion mechanism capable of actively overriding the inherent language bias of the dataset.

CONCLUSION

Conclusion & Future Work

Our investigation reveals a critical insight: the optimal depth for multimodal fusion is secondary to the foundational challenge of overcoming modality collapse. Without effectively addressing the model's inherent reliance on language priors, visual features, regardless of their integration point, will continue to be ignored.

Future Directions

Multi-Query Attention Pooling

We propose an evolution of our text-guided pooling mechanism. Instead of a single global representation, we will extract specific visual vectors for individual nouns or colors, creating multiple highly specialized visual queries. This aims to allow the decoder to focus on fine-grained visual details directly relevant to specific textual elements, rather than attempting to compress all visual information into one vector.

Bias-Mitigating Loss

To effectively break the strong language prior, our next step involves mathematical penalties. We plan to adapt bias-mitigating loss functions that actively penalize the model if it achieves a correct translation solely by relying on text, without incorporating visual input. This approach aims to mathematically enforce true multimodal dependency during training, fostering reliance on visual data without compromising valuable source information.

Thank you